

# Freight Rate Prediction

Approach, results, and lessons learned



# This is a forecasting problem

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**48,000**

training loads

**January–October 2025**

**12,000**

validation loads

**November–December 2025**

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The model predicts future months from past data. A random split does not represent that task.

# Distance is the main pricing signal

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**Freight rates are strongly related to distance and market conditions.**

A Ridge regression performed as well as more complex models in the first experiment.

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Baseline Ridge

**RMSE 633.9**

# The coordinates are redundant

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0.9995

Correlation between haversine distance and the provided distance

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**The city coordinates are synthetic.**

The provided distance was kept as the main geographic feature.

# Validation follows time

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Target encoding and label screening used training rows only.

# Log(rate) reduced percentage error

| Approach                     | RMSE         | MAPE        |
|------------------------------|--------------|-------------|
| Baseline Ridge               | 633.9        | 8.4%        |
| <b>Ridge with log target</b> | <b>633.0</b> | <b>5.8%</b> |

The target transformation improved MAPE without materially changing RMSE.

# The strongest models remain close

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| Model    | RMSE         |
|----------|--------------|
| LightGBM | <b>629.9</b> |
| CatBoost | 630.1        |
| Ridge    | 631.6        |
| XGBoost  | 632.3        |

The four models fall within 2.4 RMSE points.

# Final model

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Equal-weight average

**LightGBM + CatBoost + Ridge**

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RMSE

**629.3**

MAE

**114.3**

MAPE

**5.1%**

R<sup>2</sup>

**0.826**

The blend produced the lowest out-of-fold error.

# Two prediction files required two models

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## Validation predictions

12,000 rows

Used all available columns and the three-model average.

## December predictions

31 daily rows

Used the six shared columns and a reduced CatBoost model.

The model matched the columns available for each output.